A5 DPO Experiment Report

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1. Objective

This assignment implements Direct Preference Optimization (DPO) for human alignment on top of the GPT-2 Medium 355M SFT checkpoint. The policy model is trained with instruction-data-with-preference.json, then used to generate model_outputs.json on AlpacaEval. The generated outputs are evaluated with the course-provided Qwen judge.

Final results: raw win rate 62.61%, length-controlled win rate 82.71%, and 805 AlpacaEval examples.

2. Experimental Setup

Item	Configuration
Initial model	gpt2-355M-sft.pth
DPO model	gpt2-medium355M-dpo.pth
Model size	GPT-2 Medium 355M
Training data	instruction-data-with-preference.json
Preference examples	1100
Effective batch size	8
Optimizer	AdamW
Learning rate	1e-6
Weight decay	0.01
DPO beta	0.1
Training epochs	1 epoch
Generation strategy	Greedy decoding, max_new_tokens = 32
Evaluator	Qwen judge via AlpacaEval

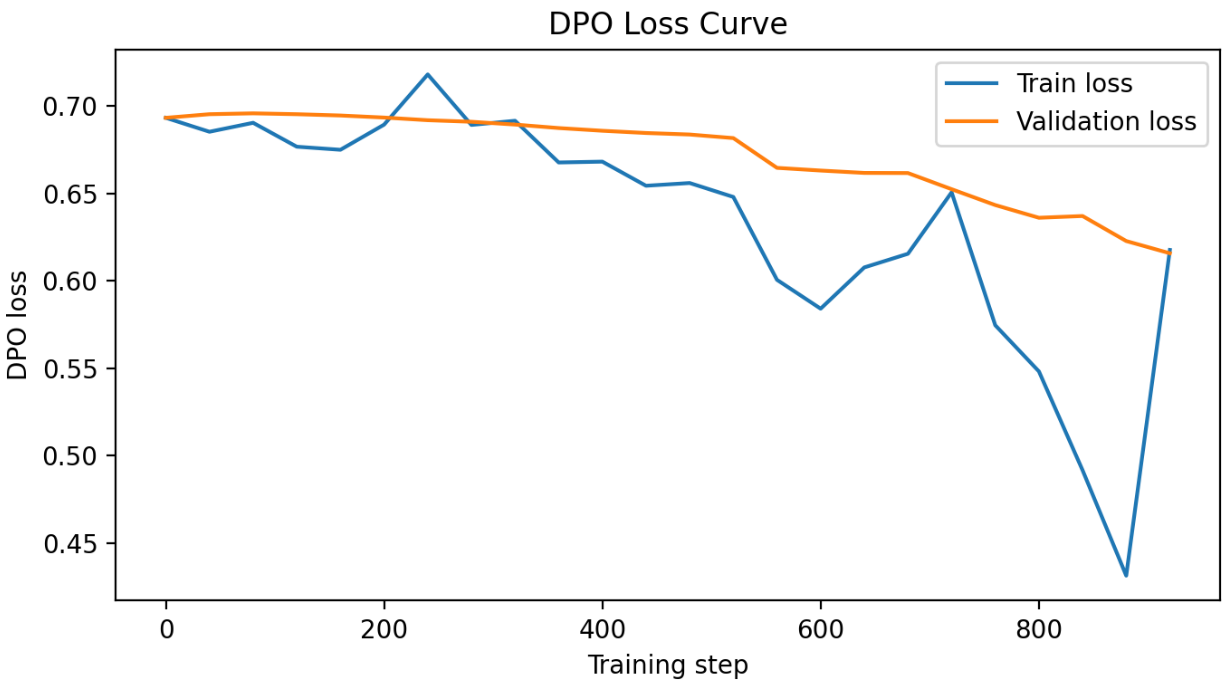
The reference model is initialized from the same SFT checkpoint as the policy model and kept frozen in evaluation mode. Its chosen and rejected response log probabilities are precomputed before policy updates. The DPO loss is computed only on response tokens; prompt tokens are excluded by the response mask.

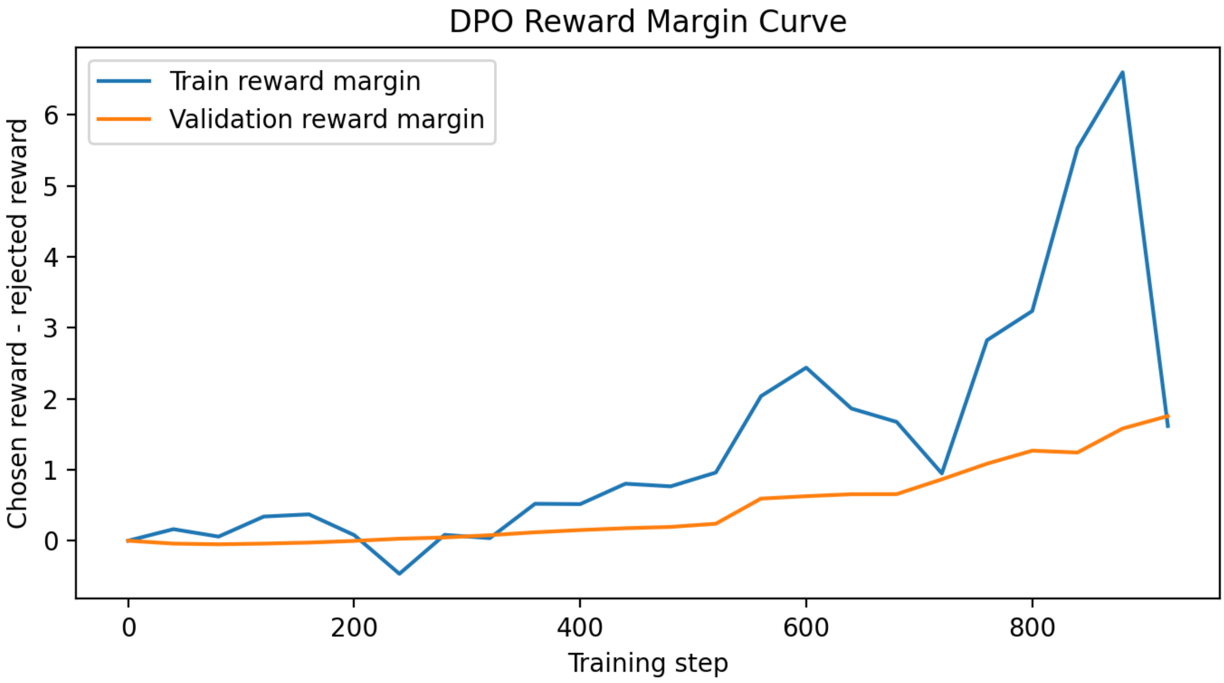
3. DPO Training Results

Metric	Initial value	Final value
Train loss	0.6931	0.6176
Validation loss	0.6931	0.6157

Metric	Initial value	Final value
Train reward margin	0.0000	1.6159
Validation reward margin	0.0000	1.7558

The loss curves decrease during training, while the reward margins increase from 0 to clearly positive values. This shows that the DPO-trained policy assigns higher relative probability to chosen responses than rejected responses compared with the frozen reference model.



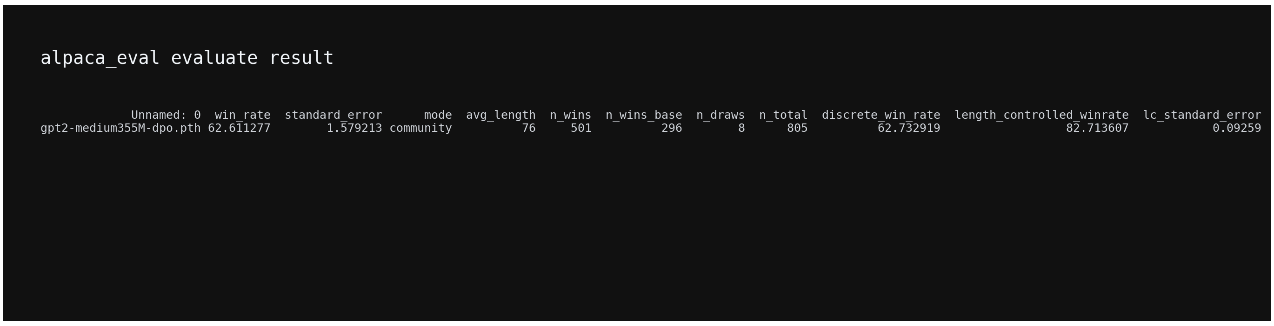


4. AlpacaEval Generation and Evaluation

The generation script generate_dpo_responses.py strictly loads the DPO checkpoint specified by --model. The final model_outputs.json contains 805 examples with an average output length of 76 characters.

The evaluation uses the course-provided Qwen judge configuration. AlpacaEval's length-controlled metric requires df_gamed.csv, so the local copy is used to avoid a HuggingFace download failure in the evaluation environment.

```
ALPACA_EVAL_DF_GAMED_PATH=/data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/a5_dpo_code/df_game
d.csv \
OPENAI_CLIENT_CONFIG_PATH=/data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/a5_dpo_code/openai_
configs.yaml \
alpaca_eval evaluate \
--model_outputs
/data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/a5_dpo_code/model_outputs.json \
--reference_outputs
/data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/a5_dpo_code/reference_outputs.json \
--annotators_config /data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/qwen_judge \
--output_path /data/student/Fengjunming/Temp/26Spring-NLP-CS310/Assignment/A5/a5_dpo_code/eval_final32_lc \
--is_overwrite_leaderboard true
```



Model	Win rate	Std err	Avg len	Wins	Ref wins	Draws	Total	Discrete	LC win
gpt2-medium355M-dpo.pth	62.61	1.58	76	501	296	8	805	62.73	82.71

5. Deliverables

The submission package contains run_dpo.py, generate_dpo_responses.py, model_outputs.json, leaderboard.csv, annotations.json, dpo_tracking.json, the loss and reward-margin curves, alpaca_eval_result.png, A5_DPO_Report.md, and Report.pdf.

The DPO checkpoint gpt2-medium355M-dpo.pth is saved locally as the resulting model. It is excluded from the submission zip because the assignment submit list requires the training script, generated model outputs, and PDF report.

6. Conclusion

This experiment completes DPO training, curve saving, model output generation, and Qwen-judge AlpacaEval evaluation. The final raw win rate is 62.61%, which is above the required 50% threshold. The length-controlled win rate is 82.71%.